

Lichen Zhu

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EDUCATION

Duke University

M.S. in Electrical and Computer Engineering

Xi'an Jiaotong-Liverpool University / University of Liverpool

B.Sc. in Financial Mathematics, First Class Honours

Durham, NC

Sep. 2025 – Jul. 2027

Suzhou, China

Sep. 2021 – Jul. 2025

EXPERIENCE

Research Intern

Sep. 2025 – Present

Duke ECE, Center for Computational Evolutionary Intelligence (CEI)

Durham, NC

- Conduct research on multimodal large language models and efficient AI systems under the guidance of Yueqian Lin, Yudong Liu, and Prof. Yiran Chen.
- Investigate efficient inference techniques for Vision-Language Models.

Research Intern

Jun. 2026 – Present

Hefei University of Technology (HFUT)

Hefei, China

- Conduct research on long-form video understanding under the guidance of Prof. Yanbin Hao and Dr. Shuo Wang.

Financial Technology Intern

Jul. 2022 – Sep. 2022

China Construction Bank

Hefei, China

- Engineered behavioral features from terabyte-scale transaction datasets using SQL and Pandas.

RESEARCH PROJECTS

Will My Assistant Remember My Allergy? What Personal LLM Assistants Forget When Conversation Memory Is Compressed

L. Zhu, Y. Lin, Y. Wang, Y. Liu, H. Li, Y. Chen

HumanSys 2026 (co-located with ACM MobiCom 2026)

- A memory-compression framework for personal LLM assistants that retains long-term user preferences while reducing storage cost.

When No Answer Is Correct: Diagnosing Absent Answer Detection for MLLMs in Video Understanding

Y. Wang, Y. Lin, L. Zhu, Y. Liu, H. Li, Y. Chen

EMNLP 2026 (arXiv:2606.08239)

- A diagnostic study showing that video MLLMs select plausible distractors instead of recognizing when the correct answer is absent, a failure that worsens under temporal reasoning and resists prompting-based fixes.

Semantic-Unit Aggregation for Streaming Interaction Memory

L. Zhu, Y. Lin, Y. Wang, H. Li, Y. Chen

Under Review, AAAI 2027

- A streaming-memory framework that organizes interaction history by semantic units rather than temporal windows, retaining original events and improving long-horizon evidence retrieval.

Query-Conditioned Evidential Keyframe Sampling for MLLM-Based Long-Form Video Understanding

Y. Wang, L. Zhu, Y. Lin, Y. Liu, J. Zhang, H. Li, Y. Chen

Under Review, WACV 2027

- An information-bottleneck formulation of keyframe selection that maximizes query-conditional mutual information and decomposes the combinatorial search into efficient frame-level scoring, improving long-form video QA under tight token budgets.

StyleVAR: Controllable Image Style Transfer via Visual Autoregressive Modeling

L. Jing, D. Zhang, P. Li, L. Zhu, Y. Xu, H. Xing

arXiv:2604.21052, 2025

- A visual-autoregressive style-transfer framework that models VQ-VAE token sequences with a scale-dependent blended cross-attention to balance style and content, outperforming AdaIN across perceptual and structural metrics.

Hybrid Deep Learning for Credit Risk Prediction

Undergraduate Dissertation, XJTLU, 2024–2025

- An XGBoost–DNN stacking ensemble for loan-default prediction over 870,000+ records, with leak-proof cross-validation and SHAP-based interpretation.

HONORS & AWARDS

Recipient of an \$18,000 scholarship from the Duke Master's Program.

2025 XJTLU Outstanding Graduate.

National Inspirational Scholarship, Ministry of Education of China (2024).

XJTLU Outstanding Student (2023–2024).

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, Pandas), SQL, R, MATLAB, Java

Foundations: Probability & Statistics, Stochastic Processes, Optimization, Time-Series Analysis, Monte Carlo Methods

Research Interests: Large Language Models (LLMs), Vision-Language Models (VLMs), Multimodal AI, Long Video Understanding, Efficient AI Systems

Languages: English (Fluent), Mandarin Chinese (Native)